



Broken glass

Brittleness

Some materials, such as window glass, are particularly brittle. They will break when pushed out of shape even a small amount.

Hardness

A scientist named Friedrich Mohs created a scale using 10 minerals to compare how hard they are. Many materials are graded on this scale.

Talc is the softest mineral



1
Talc



2
Gypsum



3
Calcite



4
Fluorite



5
Apatite



6
Feldspar



7
Quartz



8
Topaz



9
Corundum



10
Diamond

Diamond is the hardest mineral.

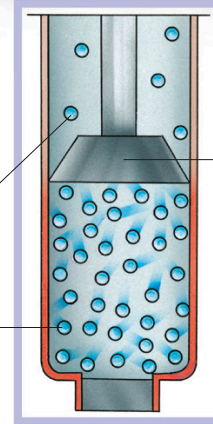
Compressibility

Gases can be squashed, or compressed, by squeezing more into the same space. This is what happens when you pump up a tire.



Gas can be compressed because its particles are far apart. A bicycle pump pushes the particles closer together.

Gas particles



Foot pump



A smooth flow

Some liquids flow more easily than others. It depends on their "stickiness," or viscosity. Hot lava from a volcano flows slowly because it is sticky.

Yes, diamond is the hardest mineral of all. It can scratch quartz.